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Market Structure, Investment, and Technical Efficiencies in Mobile Telecommunications

Contents

1 Big picture **3**

2 Data and measurement (key objects) **3**

2.1 Observed firms and products . . . 3

2.2 Demand-side data . . . 4

2.3 Infrastructure and traffic data . . . 4

2.4 Key measured objects . . . 5

2.5 Unobservables and timing . . . 5

3 Models and empirical specifications (all equations + interpretation) **5**

3.1 Consumer indirect utility and product characteristics . . . 5

3.2 Data-consumption problem . . . 6

3.3 Plan choice, nesting, market shares, and average usage . . . 6

3.4 Demand heterogeneity and modified BLP estimation . . . 7

3.5 Demand identification . . . 7

3.6 Engineering model of data transmission . . . 8

3.7 Congestion and equilibrium download speeds . . . 9

3.8 Firm pricing, infrastructure choice, and cost function . . . 9

3.9 Cost recovery from first-order conditions . . . 9

4 Identification and instruments (what makes the estimates credible) **10**

4.1 What fails in a naive approach . . . 10

4.2 What identifies the demand side in the paper's strategy . . . 10

4.3 What identifies the supply side . . . 11

4.4 Why the long-run interpretation is important . . . 11

4.5 Relation to the literature . . . 11

5 Tables and figures **12**

5.1 Table 1: Representative plan choice set 12

5.2 Table 2 and Table 3: Market shares and summary statistics 12

5.3 Figures 1 to 4: Quality, usage, and income gradients 12

5.4 Figure 5 and Table 4: Why demand identification must be indirect 14

5.5 Table 5 and Figure 6: Demand estimates and fit 14

5.6 Table 6: Cost estimates 15

5.7 Figure 7 and Figure 8: Prices, qualities, and elasticities as the number of firms changes 16

5.8 Figure 9 and Figure 10: Welfare and the optimal number of firms 16

5.9 Table 7: Symmetric versus asymmetric spectrum allocation 17

5.10 Figure 11: The social value of spectrum versus firms' willingness to pay 17

5.11 Figure 12: Short-run merger simulations 18

6 Short summary **18**

7 Conclusion **19**

7.1 Core conclusion (what the paper establishes) 19

7.2 Economic intuition 19

7.3 Takeaway 19

1 Big picture

- **Question.** How do prices, infrastructure investment, download speeds, and welfare vary with the number of mobile network operators and with the allocation of spectrum across them?
- **Setting.** The paper studies the French mobile telecommunications market in 2015, where four mobile network operators compete through national plan pricing and local infrastructure investment, and where quality is measured by delivered download speeds.
- **Core challenge.** In mobile telecommunications, more firms do not simply mean more competition. Splitting spectrum and customers across more firms can reduce prices, but it can also reduce quality because network resources are spread more thinly and queues are pooled less efficiently.
- **Main modelling contribution.** The paper combines an empirical IO demand model with an engineering model of wireless data transmission. This lets the authors map market structure into both *market power* and *technical scale efficiencies*.
- **Main theoretical insight.** There are two scale efficiencies in mobile networks:
  - **economies of density**: fewer firms imply denser customer coverage and less path loss;
  - **economies of pooling**: fewer firms pool bandwidth and stochastic demand better, which raises delivered speeds.
- **Main empirical result.** In the French market, the relevant scale gains come mostly from *pooling*, not density. More firms reduce prices, but they also reduce download speeds.
- **Main quantitative result.** In the benchmark counterfactuals, total surplus is maximized at **four firms**, while consumer surplus is maximized at about **eight firms**. High-income consumers prefer fewer firms than low-income consumers because they value faster download speeds more.
- **Spectrum result.** The marginal social value of spectrum in mobile telecommunications is much larger than firms' private willingness to pay. In the model, at four symmetric firms, the private willingness to pay is about one-fifth of the marginal consumer-surplus value of additional spectrum.
- **Merger result.** In short-run merger simulations for France, all two-firm mergers reduce consumer surplus despite operational efficiencies from combining network resources.
- **Why this paper matters.** It shows that antitrust and spectrum policy in telecom cannot be evaluated with price-only models. The right object is a market-structure model with endogenous quality and engineering-based scale economies.

2 Data and measurement (key objects)

2.1 Observed firms and products

- **Industry.** The analysis focuses on the French mobile market in October 2015.
- **Firms.** There are four mobile network operators (MNOs): Orange Mobile, SFR-Numericable, Bouygues Telecom, and Free Mobile.

- **Virtual operators.** The paper also includes the largest mobile virtual network operator (MVNO), EI Telecom, as a representative MVNO alternative.
- **Products.** The authors collect quarterly online catalogs of plans, including monthly prices, data limits, voice allowances, and contract terms.
- **Choice-set construction.** They aggregate plans into representative contracts by data-limit bracket and by whether the plan includes unlimited voice. This yields a tractable choice set of **22 representative plans**.
- **Important product characteristics.** The main vertical dimensions are price, data limit, unlimited voice, and delivered download speed. Prices and plan characteristics are set nationally; download speeds vary by operator and municipality.
- **Handset subsidies.** For plans bundled with a handset subsidy, the monthly price is adjusted by netting out the subsidy value.

2.2 Demand-side data

- **Main consumer dataset.** A proprietary Orange Mobile dataset provides the universe of residential mobile contracts in October 2015 by plan and municipality, including subscriber counts and average data usage.
- **Observed plan-level shares.** Detailed plan-by-market shares are observed only for Orange. For rival firms, the paper observes only *national firm-level shares*.
- **Quality data.** The paper uses more than one million Ookla speed tests from 2016 to measure operator-specific download speeds at the municipality level.
- **External shares.** National market shares for non-Orange firms come from GSMA Intelligence.
- **Income distribution.** Municipality-level household income deciles come from INSEE's population census.
- **Market definition.** Markets are French communes. The estimation sample keeps the 589 communes with population above 10,000, covering about 43.5% of the French population.

2.3 Infrastructure and traffic data

- **Infrastructure source.** The ANFR regulator dataset reports the location of all base stations, the number of antennas, and the frequencies operated by each network operator.
- **Bandwidth.** For each operator-market pair, the authors compute total operated spectrum.
- **Base-station geometry.** The key infrastructure object is the area served by a typical base station, summarized through an *effective cell radius*.
- **Adjusted commune area.** Because raw land area overstates the relevant area in communes with forests or other sparsely inhabited zones, the paper constructs an adjusted area measure based on population-weighted density.
- **Traffic data.** Orange's internal OSIRIS data provide total traffic volumes by network cell and over time; these data help calibrate demand rates in the congestion model.

2.4 Key measured objects

- **Aggregate shares.** In October 2015, the paper uses adjusted national shares of 29.4% for Orange, 13.4% for SFR, 17.2% for Bouygues, 21.5% for Free, 10.6% for MVNOs, and an 8.0% outside option.
- **Average quality and usage.** Across the 589 markets, average monthly data usage is 1,128 MB. Average measured download speeds are 32.82 Mbps for Orange, 23.70 for Bouygues, 23.15 for Free, 17.57 for SFR, and 24.70 for MVNOs.
- **Tariff variation.** The mean plan price is about EUR 23.47 and the mean data limit is 3,081 MB, with substantial cross-plan dispersion.
- **Effective cells.** The average effective cell radius is about 1.44 km, though the distribution is wide.

2.5 Unobservables and timing

- **Demand shocks.** Each plan-market pair has an unobserved demand component, denoted below by y_{jm}.
- **Consumer heterogeneity.** Consumers differ in price sensitivity and in their valuation of mobile data; these heterogeneities depend on income.
- **Idiosyncratic tastes.** Plan choice includes a nested-logit shock ε_{ij}.
- **Timing.** Consumers first choose a plan, then choose how much data to use on that plan. Firms simultaneously choose national prices and local infrastructure. Delivered speeds are determined jointly with usage because congestion feeds back into quality.
- **Static interpretation.** The supply model is static, but the paper interprets the recovered infrastructure cost function as a *long-run* cost object that rationalizes the observed network configuration.

3 Models and empirical specifications (all equations + interpretation)

3.1 Consumer indirect utility and product characteristics

Consumers choose both which plan to buy and how much mobile data to consume. Indirect utility from plan j in market m is

$$u_{jm}(x, \vartheta_i, \varepsilon_{ij}, v_i^p) = w_j\big(x, Q_{f(j),m}, \vartheta_i\big) + v_v v_j - v_i^p P_j + y_{jm} + \varepsilon_{ij}. \tag{1}$$

- $Q_{f(j),m}$ is the delivered download speed of the operator offering plan j in market m.
- P_j is the nationally set monthly price.
- v_j is a dummy for unlimited voice.
- y_{jm} is the plan-market demand shock.

- v_i^p is consumer i's price sensitivity.

Interpretation. This is a BLP-style utility specification, but quality here is endogenous network performance rather than a fixed product attribute.

3.2 Data-consumption problem

The utility from data use is

$$w_j(x, Q, \vartheta_i) = \vartheta_i \log(1 + x) - c_j(x, Q). \tag{2}$$

The first term captures diminishing marginal utility from data. The second term is a waiting-time cost from downloading.

With a soft data cap d_j^* and throttled speed Q^L after the cap, the download-time cost is

$$c_j(x, Q) = \begin{cases} v_c \dfrac{x}{Q}, & x \le d_j^*, \\ v_c \left(\dfrac{d_j^*}{Q} + \dfrac{x - d_j^*}{Q^L} \right), & x > d_j^*. \end{cases} \tag{3}$$

The optimal data choice is therefore piecewise:

$$x_{jm}^*(\vartheta_i) = \begin{cases} 0, & \vartheta_i \le \dfrac{v_c}{Q_{f(j),m}}, \\ \dfrac{\vartheta_i Q_{f(j),m}}{v_c} - 1, & \dfrac{v_c}{Q_{f(j),m}} < \vartheta_i < \dfrac{v_c}{Q_{f(j),m}}(d_j^* + 1), \\ d_j^*, & \dfrac{v_c}{Q_{f(j),m}}(d_j^* + 1) \le \vartheta_i < \dfrac{v_c}{Q^L}(d_j^* + 1), \\ \dfrac{\vartheta_i Q^L}{v_c} - 1, & \vartheta_i \ge \dfrac{v_c}{Q^L}(d_j^* + 1). \end{cases} \tag{4}$$

- Some consumers use no data at all.
- Some consume below the cap.
- Some bunch exactly at the cap.
- Others consume beyond the cap even after throttling.

Interpretation. The soft cap is crucial because it generates realistic bunching at the cap and realistic heterogeneity in data use across plans.

3.3 Plan choice, nesting, market shares, and average usage

Consumers do not know their exact data-value realization ex ante. Instead,

$$\vartheta_i \sim \text{Exponential}(v_i^d),$$

where v_i^d is type dependent. After integrating over the nested-logit taste shocks, the chosen plan solves

$$j_{im}^*(\varepsilon_i; h_i) = \arg\max_{j \in J \cup \{0\}} \mathbb{E}\Big[u_{jm}\big(x_j^*(\vartheta_i), \vartheta_i, \varepsilon_{ij}, v_i^p\big) \mid v_i^d \Big], \tag{5}$$

where $h_i = (v_i^p, v_i^d)$.

Conditional market shares for type h_i are

$$s_{ijm}(h_i) = \int 1\{j = j_{im}^*(\varepsilon_i; h_i)\}\, dF(\varepsilon_i), \tag{6}$$

and aggregate market shares are

$$s_{jm} = \int s_{ijm}(h_i)\, dF_m(h_i). \tag{7}$$

Average data usage among subscribers to plan j is

$$\bar{x}_{jm} = \frac{1}{s_{jm}} \iint s_{ijm}(h_i) x_{jm}^*(\vartheta_i)\, dF(\vartheta_i \mid h_i)\, dF_m(h_i). \tag{8}$$

Interpretation. The model is discrete-continuous: consumers pick a contract and then choose intensity of usage.

3.4 Demand heterogeneity and modified BLP estimation

The two main heterogeneous preference parameters vary with income z_i:

$$\begin{pmatrix} \log v_i^p \\ \log v_i^d \end{pmatrix} = \begin{pmatrix} v_{p0} \\ v_{d0} \end{pmatrix} + \begin{pmatrix} v_{pz} \\ v_{dz} \end{pmatrix} z_i. \tag{9}$$

Because detailed plan-by-market shares are only observed for Orange, the standard BLP inversion is not directly available for all firms. The paper therefore imposes, for non-Orange products,

$$y_{jm} = y_{f(j)} \qquad \text{for all } j \in J_{-\text{ORG}},$$

and develops a modified contraction mapping that rationalizes Orange plan-level shares together with national firm-level shares for the other operators.

Interpretation. This is the paper's key econometric workaround: the data are rich for one operator and coarse for the rest, so the inversion is modified rather than collapsing the whole market to one product per firm.

3.5 Demand identification

A central difficulty is that prices are national and vary very little around 2015. The paper therefore calibrates two crucial demand objects from Bourreau, Sun, and Verboven (2021):

$$e_m^{\text{ORG}}(v) = \frac{d \ln s_{\text{ORG},m}(P; v)}{d \ln P_{\text{ORG}}}, \tag{10}$$

with moment condition

$$\mathbb{E}\left[e_m^{\text{ORG}}(v)\right] = -2.36, \tag{11}$$

and the diversion ratio from Orange to the outside option,

$$\mathbb{E}\left[\text{DIV}_m^{\text{ORG},0}(v)\right] = 0.036. \tag{12}$$

The remaining parameters are identified from cross-sectional variation:

- matching predicted average usage to observed usage identifies average data valuations conditional on the download-time cost parameter v_c;
- covariance between income and usage identifies how data values vary with income (v_{dz});
- covariance between income and expensive-plan shares identifies how price sensitivity varies with income (v_{pz});
- product characteristics such as unlimited voice help identify v_v.

Because download speeds may be endogenous through congestion and firm investment, the paper instruments download speed with *log population density.* Higher density changes path loss mechanically, shifting quality independently of local demand shocks.

The moments used in GMM are summarized by Table 4 and include the calibrated Orange elasticity, the calibrated outside diversion ratio, observed average usage, the covariance of usage with income, and orthogonality moments between demand shocks and population density, data limits, and unlimited voice.

3.6 Engineering model of data transmission

The paper divides each municipality into hexagonal cells of radius R_{fm}, with one base station per cell for operator f in market m. The cell area is

$$A(R_{fm}) = \frac{3\sqrt{3}}{2} R_{fm}^2.$$

If a user at location ℓ within the cell gets local transmission rate $q_{m\ell}$, the average channel capacity per base station is the harmonic mean of these local speeds times the operated bandwidth:

$$\overline{Q}_{fm}(R_{fm}, B_{fm}) = \frac{B_{fm} A(R_{fm})}{\int_{\ell \in L(R_{fm})} q_{m\ell}^{-1}\, d\ell}. \tag{14}$$

Local transmission speed follows the Shannon-Hartley logic,

$$q_{m\ell} = g_m \log_2(1 + \mathrm{SINR}_\ell)\,, \tag{15}$$

where g_m is a market-specific spectral-efficiency term, and

$$\mathrm{SINR}_\ell(R_{fm}) = \frac{S_\ell}{N + I_\ell(R_{fm})}. \tag{16}$$

- **Economies of density.** Smaller cells shorten transmission distance and reduce path loss.
- **Economies of pooling.** Serving more consumers with pooled bandwidth and pooled stochastic demand improves queue management and increases delivered speeds.

Interpretation. The paper does not merely assume scale economies; it derives them from physical transmission laws plus queueing theory.

3.7 Congestion and equilibrium download speeds

Demand for mobile data on network f in market m is

$$X_{fm}(Q_m, P) = \text{pop}_m \sum_{j \in J_f} s_{jm}(Q_m, P)\bar{x}_{jm}(Q_m, P). \tag{18}$$

The download-request arrival rate per base station is

$$Q^D_{fm}(R_{fm}, Q_m, P) = \frac{X_{fm}(Q_m, P)}{H \cdot N_{fm}(R_{fm})}, \tag{19}$$

where H is peak downloading time in a month and $N_{fm}(R_{fm})$ is the number of base stations.

Delivered speed follows an $M/M/1$ queue logic:

$$Q_{fm} = \overline{Q}_{fm}(R_{fm}, B_{fm}) - Q^D_{fm}(R_{fm}, Q_m, P). \tag{20}$$

Interpretation. Quality is endogenous. Higher demand lowers delivered speed through congestion, and lower speed feeds back into demand.

3.8 Firm pricing, infrastructure choice, and cost function

Variable profit in market m is

$$\left(P_f - c^u_f\right) \cdot \text{pop}_m \, S^*_{fm}(P, R_m, B_m), \tag{21}$$

where c^u_f is per-user marginal cost and S^*_{fm} are equilibrium plan shares.

Infrastructure cost is

$$C_{fm}(R_{fm}, B_{fm}) = c^s_{fm} \frac{A_m}{A(R_{fm})} B_{fm}. \tag{22}$$

Market profit is

$$\Pi_{fm}(P, R_m, B_m) = \text{pop}_m \left(P_f - c^u_f\right) \cdot S^*_{fm}(P, R_m, B_m), \tag{23}$$

and national profit is

$$\Pi_f(P, R, B) = \sum_m \Pi_{mf}(P, R_m, B_m) - \sum_m C_{fm}(R_{fm}, B_{fm}). \tag{24}$$

- Firms choose national price vectors P_f and local cell radii R_f simultaneously.
- The benchmark cost function scales with operated bandwidth, which intentionally shuts down one source of fixed-cost duplication economies. This makes any measured consolidation gain come mostly from transmission and congestion rather than accounting conventions.

3.9 Cost recovery from first-order conditions

Per-user costs are backed out from the pricing FOCs. The first-order condition is

$$\left[\sum_m \text{pop}_m S^*_{fm}(P, R_m, B_m) + \sum_m \text{pop}_m J_f S^*_{mf}(P, R_m, B_m)\right] \left(P_f - c^u_f\right) = 0, \tag{25}$$

which yields the cost inversion

$$\hat{c}_f^u = P_f + \left[\sum_m \text{pop}_m J_f S_{mf}^*(P, R_m, B_m)\right]^{-1} \sum_m \text{pop}_m S_{mf}^*(P, R_m, B_m). \tag{26}$$

Infrastructure costs are recovered by matching marginal operating income from reducing the cell radius to marginal infrastructure cost. Numerically, marginal revenue is approximated by

$$MR_{fm}^R(R_m, B_m) = \frac{\Pi_{fm}(P, R_{fm} + 0.01, R_{-f,m}, B_m) - \Pi_{fm}(P, R_{fm} - 0.01, R_{-f,m}, B_m)}{0.02}. \tag{27}$$

Interpretation. Demand estimates plus optimal pricing and investment conditions let the paper infer both customer-level variable cost and base-station cost without directly observing firm cost data.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in a naive approach

- **Little price variation.** Prices are set nationally and barely change over the two years before the sample period, so standard demand estimation from price variation is weak.
- **Partial market-share observation.** Product-level shares are only observed for Orange, so a standard full-market BLP inversion cannot be used directly.
- **Endogenous quality.** Download speeds depend on both investment and congestion, so observed quality is potentially correlated with demand shocks.
- **Short-run switching concern.** Mobile plans often involve long contract durations, so short-run share responses could understate long-run substitution.

4.2 What identifies the demand side in the paper's strategy

- **External elasticity calibration.** The Orange own-price elasticity and diversion to the outside option are anchored to an earlier structural study with richer price variation.
- **Cross-sectional quality variation.** Variation in download speeds across communes identifies the value consumers place on quality and data usage.
- **Population density instrument.** Log population density shifts quality through path loss and network geometry, helping separate quality from demand shocks.
- **Income heterogeneity.** Richer markets sort more into expensive and high-data plans, which identifies heterogeneity in price sensitivity and data valuation.
- **Modified BLP inversion.** The assumption of common non-Orange demand shocks within firm lets the paper rationalize Orange plan shares and rival firm shares simultaneously.

TABLE 1
Choice Set

Operator, Price (€)	Data Limit (MB)	Unlimited Voice	Plans Represented	Min Price (€)	Max Price (€)	Min Limit (MB)	Max Limit (MB)
ORG:							
12.07	50	No	11	4.99	30.99	0	50
14.99	1,000	No	4	14.99	14.99	1,000	1,000
22.91	1,000	Yes	2	22.91	24.99	1,000	1,000
30.91	4,000	Yes	5	19.99	48.99	3,000	5,000
38.74	8,000	Yes	11	38.74	165.99	8,000	20,000
BYG:							
8.07	0	No	6	3.99	11.32	0	20
14.99	1,000	No	3	14.99	14.99	1,000	1,000
20.91	3,000	Yes	4	19.99	29.99	3,000	5,000
33.74	10,000	Yes	4	32.70	72.70	10,000	20,000
FREE:							
2.00	50	No	1	2.00	2.00	50	50
19.99	3,000	Yes	1	19.99	19.99	3,000	3,000
SFR:							
12.07	100	No	5	5.99	14.99	100	200
14.99	1,000	No	3	14.99	19.99	1,000	1,000
22.91	1,000	Yes	3	22.91	29.99	1,000	1,000
31.91	5,000	Yes	5	19.99	43.99	3,000	5,000
37.74	10,000	Yes	9	36.70	149.99	10,000	20,000
MVNO:							
7.99	0	No	13	7.99	18.99	0	200
17.99	1,000	No	5	9.99	17.99	500	1,000
19.99	500	Yes	10	19.99	35.99	500	2,000
42.99	5,000	Yes	13	12.99	61.99	3,000	5,000
64.99	10,000	Yes	4	64.99	76.99	10,000	10,000

Note.—Each row corresponds to an object in the choice set, i.e., a representative product. The minimum and maximum prices and data limits are over the set of plans represented by each representative product in the choice set. The representative plan is the least expensive within the group of plans that has a commitment of 24 months; therefore, because some plans have commitments of less than 24 months, it is possible for the minimum price to be lower than the representative plan's price. ORG = Orange Mobile; BYG = Bouygues Telecom; SFR = SFR-Numericable; FREE = Free Mobile.

4.3 What identifies the supply side

- **Pricing FOCs** identify per-user marginal costs once demand and elasticities are known.
- **Infrastructure FOCs** identify per-base-station costs by matching marginal revenue from better quality to marginal investment cost.
- **Engineering discipline** is crucial: without the physics-based mapping from base stations and bandwidth into channel capacity, the infrastructure side would be underidentified.

4.4 Why the long-run interpretation is important

- The estimation leans on cross-sectional variation in market shares and quality rather than high-frequency price changes.
- This makes the recovered preferences and infrastructure costs closer to long-run trade-offs than to short-run locked-in behavior under mobile contracts.

4.5 Relation to the literature

- On the demand side, the paper extends BLP-style discrete-choice demand to a setting with endogenous continuous usage and endogenous quality.
- On the supply side, it contributes to the literature on competition and investment by showing that investment must be evaluated through delivered service quality, not only through raw spending.
- Relative to standard merger analysis, the paper highlights that telecom is a sector where engineering-based scale efficiencies can materially offset or amplify price effects.

5 Tables and figures

5.1 Table 1: Representative plan choice set

Read:

- The paper compresses the rich French menu of mobile contracts into **22 representative plans** across MNOs and a representative MVNO.
- The key dimensions preserved are price, data allowance, and unlimited voice, because these are the main margins of differentiation in 2015.
- This table matters because all later structural results are driven by how consumers sort across these representative vertical tiers.

5.2 Table 2 and Table 3: Market shares and summary statistics

TABLE 2
Aggregate Market Shares of Alternatives

Market Size (Millions)	ORG	SFR	BYG	FREE	MVNO	Nonusers
56.5	29.4%	13.4%	17.2%	21.5%	10.6%	8.0%

Note.—Data reported by the regulator (ARCEP 2016) provide the relative share of MVNOs and MNOs. Relative shares within MNOs are obtained from GSMA Intelligence. Shares are adjusted to allow for 8% outside option share, consistent with CREDOC (2015).

TABLE 3
Summary Statistics

	Mean	SD	Minimum	Maximum
	A. Quality and Market Data			
Market average usage (MB)	1,128	204	579	1,829
Quality ORG (Mbps)	32.82	11.11	3.97	84.98
Quality BYG (Mbps)	23.70	9.65	.60	72.97
Quality FREE (Mbps)	23.15	11.03	1.56	56.74
Quality SFR (Mbps)	17.57	8.58	.39	52.30
Quality MVNO (Mbps)	24.70	7.04	5.13	48.87
Median income (€)	13,035	3,177	5,152	31,320
Markets (n)	589			
	B. Tariff Data			
Price	23.47	14.22	2.00	64.99
Price (ORG)	23.92	9.90	12.07	38.74
Price (others)	23.33	15.32	2.00	64.99
Data limit	3,081	3,484	0	10,000
Number of phone plans	22			
	C. Infrastructure Data			
Bandwidth per firm (MHz)	70.69	30.42	.00	140.20
Base stations (n)	7.47	21.47	0	511
Effective cell radius (km)	1.44	.93	.26	7.64

Note.—Customer, quality, market, and infrastructure data summary statistics are (unweighted) across markets. Tariff data summary statistics are across mobile phone plans.

Read:

- The market is sizable and competitive: 56.5 million potential consumers, with Orange at 29.4%, Free at 21.5%, Bouygues at 17.2%, SFR at 13.4%, MVNOs at 10.6%, and an 8% outside option.
- Quality differences are large: average speeds range from about 17.6 Mbps for SFR to 32.8 Mbps for Orange across the sample communes.
- Infrastructure heterogeneity is also large, with effective cell radii spanning from about 0.26 km to 7.64 km. This is the raw material that identifies quality production.

5.3 Figures 1 to 4: Quality, usage, and income gradients

Read:

- **Figure 1** shows broad dispersion in average measured download speeds across markets and operators.

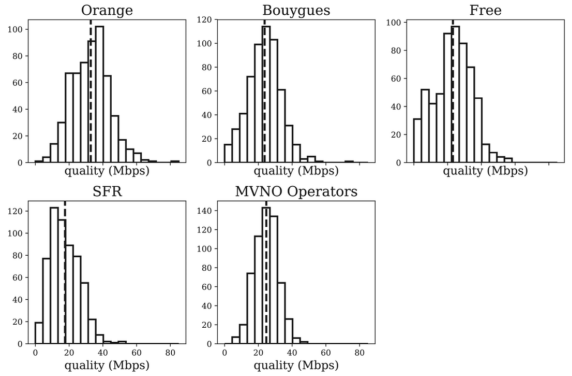

FIG. 1.—Histograms of qualities by operator. The figure shows the average download speed at the market level for each operator. Dashed vertical lines represent the average of the market-level average by operator, unweighted by population. The scale of the *x*-axis is the same across all subplots, allowing for comparisons in the distribution of average download speeds across operators.

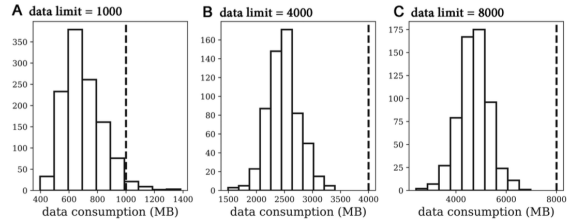

FIG. 3.—Average data usage across markets. Each subplot represents a histogram of customers' average data consumption at the market level for Orange phone plans with a particular data limit. Market-level average data consumption for each phone plan is obtained by, for each market, averaging data consumption across customers in that market subscribing to the phone plan. Dashed vertical lines represent the average of the market-level averages, unweighted by population.

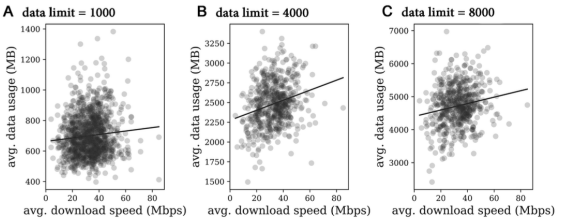

FIG. 2.—Average data usage versus measured quality across markets. The figure presents scatterplots displaying the relationship between the average download speed and the average amount of data consumed per customer for three different Orange phone plans. Observations are at the market level. The line in each subplot is a line of best fit for the observations.

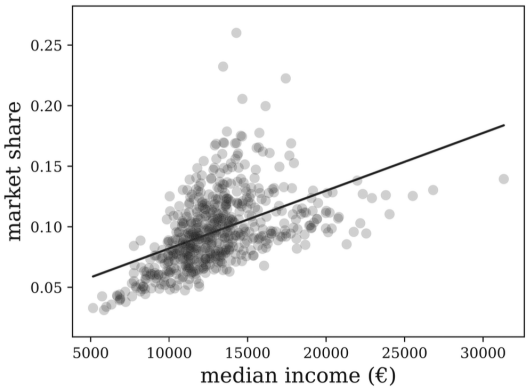

FIG. 4.—Median income versus expensive contract market share. The figure presents a scatterplot of the median income in a market against the market share of Orange's two most expensive phone plans (4 and 8 GB representative plans). The line is a line of best fit.

- **Figure 2** shows that faster networks are associated with greater average data use within a plan tier, which is exactly the demand-side channel the model exploits.
- **Figure 3** shows that average usage is well below plan caps for most plans, so data limits matter mostly through anticipated flexibility rather than constant binding.
- **Figure 4** shows richer markets buy more expensive, high-data plans, providing the income gradient that identifies heterogeneous willingness to pay.

5.4 Figure 5 and Table 4: Why demand identification must be indirect

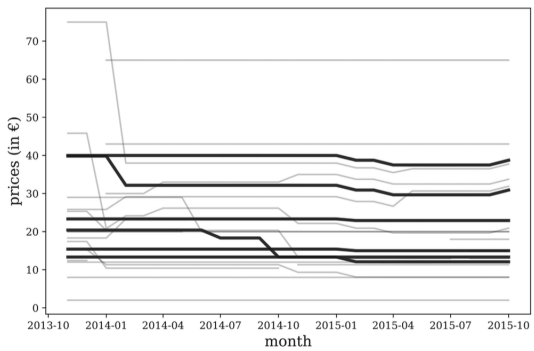

FIG. 5.—Prices of Orange phone plans (black lines) and other MNOs' phone plans (gray lines) over 2 years.

TABLE 4
MOMENTS

$\mathbb{E}[-e_m^{\text{ORG}}(\theta) - 2.36] = 0$
$\mathbb{E}[\text{DIV}_m^{\text{ORG},0}(\theta) - .036] = 0$
$\mathbb{E}[(\xi_{jm}(\theta) - \theta_{\text{O}})\text{inc}_m^{\text{med}}] = 0$
$\mathbb{E}[\bar{x}_{jm}(\theta) - \bar{x}_{jm}] = 0$
$\mathbb{E}[(\bar{x}_{jm}(\theta) - \bar{x}_{jm})\text{inc}_m^{\text{med}}] = 0$
$\mathbb{E}[(\xi_{jm}(\theta) - \theta_{\text{O}})\log(\text{pop_density}_m)] = 0$
$\mathbb{E}[(\xi_{jm}(\theta) - \theta_{\text{O}})\bar{d}_j] = 0$
$\mathbb{E}[(\xi_{jm}(\theta) - \theta_{\text{O}})v_j] = 0$
$\mathbb{E}[\xi_{jm}(\theta) - \theta_{\text{O}}] = 0$

Read:

- **Figure 5** shows almost no time-series price variation in the two years before the sample. So a standard price-based demand estimation strategy is not available.
- **Table 4** shows the GMM moments: calibrated Orange elasticity and outside diversion, plus moments using usage, income, population density, data limits, and unlimited voice.
- The identification message is that the paper substitutes *cross-sectional variation plus external elasticity information* for the missing time-series price variation.

5.5 Table 5 and Figure 6: Demand estimates and fit

Read:

- Price sensitivity declines with income, while the parameter governing the distribution of data utility rises with income.
- Willingness to pay rises strongly with income. For example, the willingness to pay to move from a 1-GB to a 4-GB plan rises from about EUR 2.73 at the 10th income percentile to EUR 4.77 at the 90th percentile; willingness to pay for unlimited voice rises from EUR 3.81 to EUR 20.19; willingness to pay to raise speed from 10 to 20 Mbps rises from EUR 1.90 to EUR 4.68.
- **Figure 6** shows the model matches average Orange usage quite well across low-, medium-, and high-data plans.

TABLE 5
Demand Parameter Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	A. Parameter Estimates							
	$\hat{\theta}_{p0}$	$\hat{\theta}_{pc}$	$\hat{\theta}_{v}$	$\hat{\theta}_{O}$	$\hat{\theta}_{d0}$	$\hat{\theta}_{dc}$	$\hat{\theta}_{\epsilon}$	$\hat{\sigma}$
Estimates	−1.859	−.727	.460	2.375	.597	.335	1.405E–4	.683
	(.687)	(.221)	(.180)	(.272)	(.380)	(.039)	(4.108E–5)	(.111)
	B. Willingness to Pay (€)							
	Percentile							
	10th	30th	50th	70th	90th			
1-GB → 4-GB plan	2.73	3.16	3.52	3.95	4.77			
Unlimited voice	3.81	5.53	7.44	10.43	20.19			
10 Mbps → 20 Mbps	1.90	2.39	2.84	3.41	4.68			

Note.—Rather than estimating θ_ϵ and σ directly, we estimate $\log(\theta_\epsilon)$ and $\log(\sigma/(1-\sigma))$ since $\theta_\epsilon > 0$ and $0 < \sigma < 1$, respectively. We derive the associated standard errors using the delta method. The estimates for θ_ϵ are in scientific notation. Measures of willingness to pay implied by the demand parameter estimates are provided in panel B. Willingness to pay for upgrading from a 1-GB to a 4-GB plan is calculated holding download speeds fixed at the median speed observed in the data. Willingness to pay for upgrading from a download speed of 10 Mbps to 20 Mbps is calculated for a 10-GB phone plan.

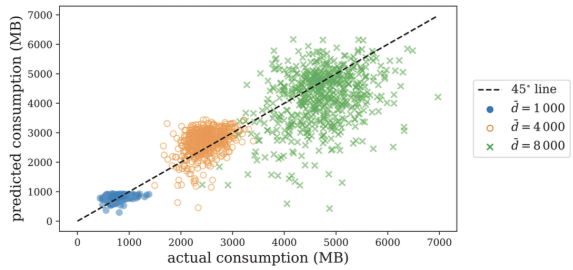

Fig. 6.—Actual average data consumption versus the consumption predicted by our demand parameter estimates for three Orange phone plans. Dashed line is a 45° line.

TABLE 6
Cost Estimates

	(1)	(2)	(3)	(4)
	A. Data Limit Range			
	$\bar{d} < 1{,}000$	$1{,}000 \leq \bar{d} < 5{,}000$	$\bar{d} \geq 5{,}000$	
Per-user costs $\hat{c}_j^u$ (€)	4.95	10.33	20.53	
	(.65)	(.66)	(2.02)	
	B. MNO			
	ORG	SFR	FREE	BYG
Per–base station costs $\hat{C}_f$ (€)	182,197	140,556	142,792	201,733
	(60,698)	(40,035)	(46,587)	(67,896)

Note.—Per-user cost values represent the average estimated cost per user, calculated across all products within the data limit range specified. Values in parentheses are the average standard errors. Per–base station costs are in long-run per–base station terms (vs. monthly per–base station units of bandwidth terms, as introduced in the text). To create long-run per–base station costs, we first use the monthly per–base station units of bandwidth costs that we recover from our estimates and multiply them by 75. This corresponds to the estimated monthly cost of a base station operating 75 MHz of bandwidth, which corresponds to the average amount of bandwidth per firm across markets. To recover the cost of long-lived base stations from our estimates based on monthly profits, we assume a monthly discount rate of 0.5%. The above results are therefore 201 times the per–base station costs we recover. Values in parentheses are standard deviations of the distribution of estimated costs across markets (not standard errors of the estimates).

- The economic point is that richer households buy better plans not because they intrinsically value data more in utility terms, but because they are less price sensitive and therefore pay more for better quality.

5.6 Table 6: Cost estimates

Read:

- Estimated per-user costs rise sharply with the data tier: about EUR 4.95 for low-data plans, EUR 10.33 for medium-data plans, and EUR 20.53 for high-data plans.
- Estimated long-run per-base-station costs range from roughly EUR 140,000 to EUR 202,000 across operators.
- These values line up well with external engineering and industry benchmarks, which supports the supply-side inversion.

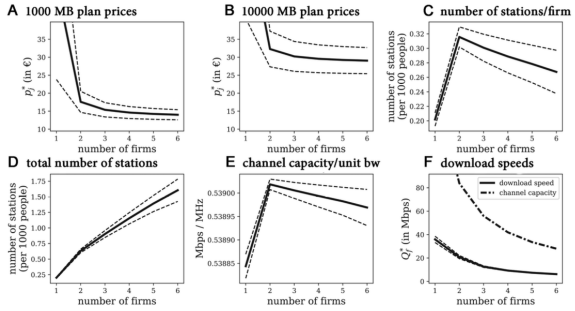

Fig. 7.—Counterfactual prices and qualities. Channel capacity is per base station. Download speeds are the average speed of transmission received by a user, including wait times. Dashed lines represent 95% confidence intervals.

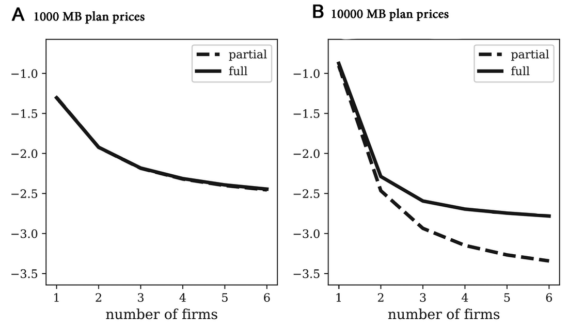

Fig. 8.—Full and partial price elasticities for two types of phone plans. Partial elasticities are derivatives in which download speeds are held fixed. Full elasticities take into account how download speeds change endogenously as prices change. Price elasticities are evaluated at the equilibrium prices and quantities.

5.7 Figure 7 and Figure 8: Prices, qualities, and elasticities as the number of firms changes

Read:

- **Figure 7** shows the core telecom trade-off: more firms reduce prices, but they also reduce delivered download speeds.
- Investment per firm is nonmonotonic, but quality still declines with the number of firms because total bandwidth and demand pooling are split more thinly across operators.
- **Figure 8** shows why telecom pricing differs from standard differentiated-products logic: full price elasticities account for endogenous congestion, so they are smaller in magnitude than partial elasticities that hold quality fixed.
- The fact that full elasticities decline less with the number of firms means that quality feedback dampens the price competition one would otherwise predict.

5.8 Figure 9 and Figure 10: Welfare and the optimal number of firms

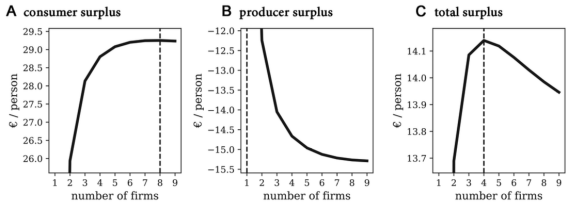

Fig. 9.—Counterfactual welfare. The figure displays measures of welfare as a function of the number of firms. Welfare is measured in euros per capita relative to the monopoly case, so for each plot, the welfare value at one firm is 0.Dashed vertical lines indicate the number of firms that maximizes each measure of welfare.

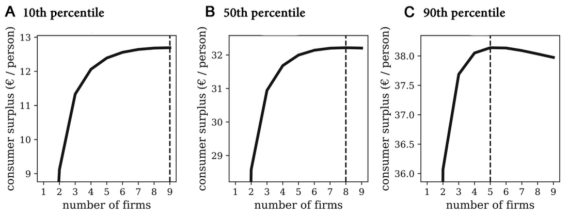

Fig. 10.—Counterfactual welfare by income level. Welfare is measured in euros per capita relative to the case of monopoly. Dashed vertical lines indicate the number of firms that maximizes that measure of welfare.

Read:

- Total surplus is maximized at about **four firms**, while consumer surplus is maximized at about **eight firms**.
- **Figure 10** shows strong heterogeneity by income: low-income consumers want more firms because they care more about lower prices, while high-income consumers prefer fewer firms because they value speed more.

- The welfare-maximizing industry structure is therefore criterion dependent: antitrust based on consumer surplus alone points to more fragmentation than total-surplus analysis does.

5.9 Table 7: Symmetric versus asymmetric spectrum allocation

Read:

- The paper compares a symmetric three-firm spectrum allocation to an asymmetric one in which one firm gets one-half of total bandwidth and the other two firms get one-quarter each.
- The firm with more spectrum offers faster download speeds and slightly higher prices, while the two smaller-spectrum firms offer slower speeds and lower prices.
- But the asymmetric allocation is still inefficient overall: it lowers consumer surplus by about EUR 0.03 per person and lowers total surplus by about EUR 0.02 per person relative to the symmetric three-firm allocation.
- So even if concentration across firms can raise private profits, asymmetric spectrum holdings are not socially desirable in this environment.

TABLE 7
Three-Firm Equilibria under Different Spectrum Allocations

	(1)	(2)	(3)	(4)
	A. Symmetric and Asymmetric Three-Firm Allocations			
	Firm Bandwith Share	1,000 MB Plan Price (€)	10,000 MB Plan Price (€)	Download Speed (Mbps)
Symmetric allocation:				
Equal allocation firm	$(1/3)B$	15.377	30.222	12.589
		(1.010)	(2.120)	(.317)
Asymmetric allocation:				
Large allocation firm	$(1/2)B$	15.498	30.313	15.473
		(1.059)	(2.149)	(.491)
Small allocation firm	$(1/4)B$	15.319	30.187	10.949
		(.984)	(2.103)	(.237)
	B. Surplus Relative to Four-Firm Symmetric Allocation			
	Δ Consumer Surplus (€/person)	Δ Producer Surplus (€/person)	Δ Total Surplus (€/person)	
Symmetric allocation	−.668	.614	−.054	
	(.154)	(.118)	(.044)	
Asymmetric allocation	−.701	.625	−.076	
	(.161)	(.122)	(.047)	
Difference	.032	−.010	.022	
	(.007)	(.003)	(.004)	

Note.—Rows correspond to firms' phone plan characteristics under either a symmetric spectrum allocation or an asymmetric one (for which there are two types of firms, one with one-half of the total bandwidth and two with one-quarter of the total bandwidth). Panel B presents the surplus relative to the four-firm symmetric case. The final row presents the difference between the symmetric and asymmetric values.

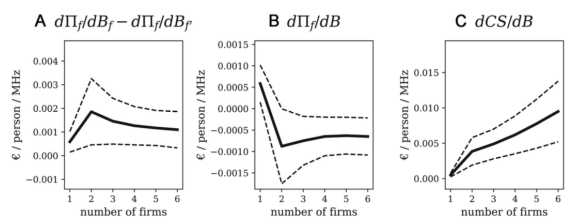

Fig. 11.—Bandwidth derivatives evaluated at symmetric equilibrium values. Values are rates per month. The derivative of own profits with respect to another firm's bandwidth ($d\Pi_f/dB_{f'}$) is undefined in the monopoly case. Therefore, in *A*, what is reported in the case of only one firm is simply the derivative of own profits with respect to own bandwidth ($d\Pi_f/dB_f$). Dashed lines represent 95% confidence intervals.

5.10 Figure 11: The social value of spectrum versus firms' willingness to pay

Read:

- At four symmetric firms, the model implies a private willingness to pay for spectrum of about EUR 0.25 per person per MHz, while the marginal consumer-surplus value is about EUR 1.24 per person per MHz.

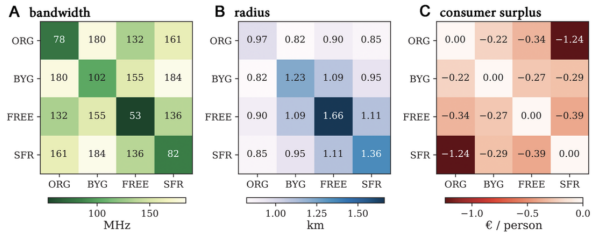

FIG. 12.—Short-run counterfactual consumer surplus under mergers. MNOs are displayed along the *x*- and *y*-axes. Points correspond to the merger of the MNOs along the *x*- and *y*-axes, and the diagonal corresponds to no merger. *A*, *B*, Bandwidths (*A*) and radii (*B*) of the merged and nonmerged MNOs. Values along the diagonal are population-weighted averages of the bandwidth or radius of the MNO. Off-diagonal values are the population-weighted averages of the bandwidth or radius of the merged MNO (while for nonmerged firms in that merger scenario, the values remain the same as the ones for those MNOs along the diagonal). *C*, Change in consumer surplus (relative to the no-merger case) for the different merger scenarios.

- The 2015 French 700 MHz auction raised about EUR 0.70 per person per MHz, which also lies below the model-implied social value.
- The key point is that auction bids reveal private surplus, not the full social value of spectrum in a quality-constrained industry.

5.11 Figure 12: Short-run merger simulations

Read:

- The paper simulates short-run mergers that combine ownership and some network resources while keeping product sets and infrastructure largely fixed.
- Every possible two-firm merger reduces consumer surplus.
- The losses range from about EUR 0.22 per person per month (Orange-Bouygues) to about EUR 1.24 per person per month (Orange-SFR).
- The message is sharp: even though mergers generate pooling efficiencies, short-run market-power effects dominate in the French data.

6 Short summary

- The paper builds a structural model of mobile telecommunications in which quality is endogenous and generated by infrastructure, congestion, and spectrum.
- The big economic trade-off is simple: more firms imply more price competition but lower download speeds because scale economies in pooling are lost.
- In the calibrated French market, consumer surplus prefers more firms than total surplus does, asymmetric spectrum allocations are inefficient, spectrum auctions understate the social value of spectrum, and short-run mergers reduce consumer welfare.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Mobile telecommunications is a setting where competition policy and engineering constraints must be analyzed jointly.
- The paper shows that it is possible to estimate such a joint model structurally using consumer plan-choice data, measured download speeds, and infrastructure data.
- The resulting counterfactuals show that market structure affects not only prices but also the quality of service, the value of spectrum, and the welfare consequences of mergers.

7.2 Economic intuition

- **Price effect.** More firms intensify competition and push prices down.
- **Quality effect.** But more firms also fragment bandwidth and customer pools, which lowers delivered speeds.
- **Distributional effect.** Richer consumers care more about speed and therefore tolerate more concentration; poorer consumers care more about price and prefer more entry.
- **Policy effect.** Since firms internalize only private profit, they bid too little for spectrum relative to its consumer-surplus value, and merger analysis based on prices alone misses the central quality channel.

7.3 Takeaway

This paper is best understood as a telecom-specific generalization of differentiated-products competition: instead of treating quality as fixed or directly chosen, it derives quality from physics, congestion, and infrastructure. That move changes the policy conclusions. In the French mobile market, more competition is not unambiguously better, because competition also degrades service quality. At the same time, the model does not give a free pass to consolidation: short-run mergers still hurt consumers, and asymmetric spectrum allocations are inefficient. The broader lesson is that in regulated infrastructure industries, antitrust, investment, and spectrum policy have to be studied inside one unified equilibrium framework.